Manya Wang

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EDUCATION

National University of Singapore — **Singapore**
B.Sc. in Data Science & Analytics; Second Major: Quantitative Finance | GPA: 4.2/5.0 — *Aug 2024 – Jun 2028*

- Coursework: Machine Learning, Mathematical Statistics, Data Structures & Algorithms, Convex Optimization

Wuhan No.6 High School, International Department — **Wuhan, China**
Advanced Placement Program — *Sep 2021 – Jun 2024*

TECHNICAL SKILLS

- **Programming:** Python, R, SQL
- **Machine Learning / Deep Learning:** PyTorch, scikit-learn; model training, fine-tuning and evaluation
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn; EDA, feature engineering, statistical modeling
- **Development & Deployment:** Git, Streamlit, Streamlit Cloud, Label Studio, Codex, Claude Code
- **Certifications:** MIIT Large Model Application Development Engineer (Junior), Jun 2026

PROJECT EXPERIENCE

AI-Based Computer Vision Safety Monitoring System — **Aura International AI Program**
Computer Vision Project under Prof. Xiaohua Li — *Aug 2025*

- Expanded the training set 4× to ~10.5k images by unifying labels from multiple sources into YOLO format, fixing invalid annotations and adding smoking data, giving the 7-person team a clean dataset for training
- Fine-tuned a pretrained YOLOv8s model and benchmarked it against YOLOv8n and YOLOv8m, raising mAP@0.5 from 0.797 to 0.861 and F1 from 0.74 to 0.81 over the baseline
- Diagnosed failure modes using confusion matrices and PR/F1 curves, identifying smoking false positives as the main error source, and presented the findings in the team's final report and slides

Fashion-MNIST Image Classification & Web Deployment — **NUS, Singapore**
Machine Learning Project — *Mar 2026 – Apr 2026*

- Benchmarked logistic regression, SVM and random forest with and without PCA dimensionality reduction, identifying random forest as the strongest classical model at 87.65% test accuracy
- Built a PyTorch inference pipeline and Streamlit app around the 6-person team's CNN (~93% test accuracy), enabling real-time classification of uploaded images
- Deployed the app on Streamlit Cloud by fixing dependency and path issues, and diagnosed its failures on real-world photos as domain shift and preprocessing mismatch

Singapore HDB Resale Price Modeling — **NUS, Singapore**
Data Analytics & Statistical Modeling Project — *Apr 2026 – May 2026*

- Engineered region, storey and remaining-lease features from 2,456 HDB resale records in R to quantify price drivers
- Diagnosed multicollinearity between floor area and flat type using VIF, then compared ridge regression with stepwise AIC selection to build a log-linear OLS model explaining ~81% of price variation (adjusted $R^2 \approx 0.81$)

RESEARCH EXPERIENCE

Zero-Knowledge Proofs and Epistemic Logic Research — **Online**
Research Student under Prof. Joseph Y. Halpern, Cornell University — *Jun 2023 – Oct 2023*

- Authored a paper, accepted at CONF-CIAP 2024, applying epistemic logic and Kripke structures to analyze the definitions and properties of zero-knowledge proofs, illustrated with representative protocols built on modular arithmetic and RSA

WORK EXPERIENCE

Avantclass — **Wuhan**
AP & Competition Tutor — *Sep 2024 – May 2026*

- Delivered 1,000+ hours of one-on-one instruction in AP Calculus, Physics, Computer Science and Statistics, with lesson plans and exam strategies tailored to each student
- Achieved a 100% score-of-5 rate among students who sat AP exams; coached Physics Bowl and British Physics Olympiad candidates

ADDITIONAL INFORMATION

- **Languages**: Mandarin (Native), English (Fluent; TOEFL 110)
- **Interests**: Sudoku, Card Games, Puzzle Games